Software Requirements Specification for Flexin: subtitle describing software

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

1 Purpose of the Project

1.1 User Business

Insert your content here.

1.2 Goals of the Project

Insert your content here.

2 Stakeholders

This section will look at all individuals and groups that will affected by the success of this project. We will look further and study the people affected through the following sections.

2.1 Client

The client with whom the team is building the app is **Chris Schankula**. Chris was responsible for providing the initial idea and will work throughout each stage of the project to deliver the final product. *Insert your content here.*

2.2 Customer

- **Primary**: Young adults aged 18-30 who are seeking to build financial literacy and sustainable fitness habits.
- **Secondary:** Adults older than the intended age range who are looking to learn new skills, reinforce or start new healthy habits.

Insert your content here.

2.3 Other Stakeholders

- **Financial Institution/Banks:** which benefit from financially literate consumers who will engage more with their finances.

- **Gyms/Training Facilities:** which can benefit from increased user activity by providing a place to exercise.

Insert your content here.

2.4 Hands-On Users of the Project

- **Mobile (Android) Smartphone Users:** users will be able to interact with the app and utilize the different features to view and track financial and fitness goals. For the purpose of Capstone our final deployed app will be for Android users only.

Insert your content here.

2.5 Personas

2.5.1 Brandon - The Novice Saver and Avid Runner

Background

Brandon is a university student in his final year of Environmental Sciences at the University of Toronto, preparing for graduation in the spring. Brandon has recently begun learning more about his own finances, but still has a limited understanding of investing and saving.

Goals and Motivations

Brandon enjoys recreational running as a way to stay active, relieve stress, and maintain a healthy lifestyle. He wants to improve running performance, build consistency, and take this hobby to the next level. At the same time, Brandon wants to develop his financial literacy and gain confidence in investment decisions before investing real savings.

Behaviours and Habits

Brandon balances school responsibilities with running and personal development. He currently uses his phone as his main device to track his running activities such as goals and progress. Brandon prefers simple explanations, interactive learning, and measurable progress rather than complicated financial terminology.

Needs and Expectations

- A risk-free stock trading simulator that uses virtual money.
- Beginner-friendly lessons explaining investment opportunities, market trends, risk, and investment strategies.
- Progress tracking and feedback to help build investing knowledge and confidence.
- Running features to record workouts, monitor distance and pace, and track improvements over time.
- Achievable goals, milestones, and challenges that encourage consistent participation in both activities.

Frustrations

Brandon finds financial platforms intimidating because of unfamiliar terminology and the potential to lose money. Many investing resources assume prior knowledge, while running apps may focus primarily on recording workouts rather than helping beginners improve. Brandon wants a straightforward experience that makes learning and progress rewarding.

2.5.2 Connor – The Young Expert

Background

Connor is a third-year Commerce student specializing in Finance, returning to university after completing a year-long co-op placement at IG Wealth Management. Coming from a long line of financial professionals, Connor has developed a strong interest in financial management and aspires to build a successful career in the industry. As a top-performing student and hard-working individual, Connor is confident in his financial knowledge and enjoys applying what he has learned to real-world situations.

Goals and Motivations

Connor wants to excel academically and build on the professional experience gained during his co-op placement to advance his career in financial management. He enjoys demonstrating his financial knowledge and sharing his

opinions on investment opportunities and market trends. Although Connor is an avid sports fan, he recognizes that balancing academics, career ambitions, and personal interests leaves him with little time to exercise. He wants to find ways to incorporate physical activity into his busy schedule without compromising his professional goals.

Behaviours and Habits

Connor maintains a demanding schedule focused on academics, career development, and expanding his financial knowledge. He regularly follows financial news, monitors market trends, and enjoys discussing investment strategies with others. Connor is competitive and motivated by measurable results, academic achievement, and opportunities to demonstrate his expertise. While he enjoys watching sports, exercise is not currently a consistent part of his routine due to his time commitments.

Needs and Expectations

- Advanced financial tools and simulated trading features that allow him to apply his existing investment knowledge.
- Access to market trends, investment opportunities, and performance statistics to support informed decision-making.
- Competitive features such as rankings, challenges, and performance comparisons to showcase his financial expertise.
- Efficient running and exercise tracking that fits into his busy academic and professional schedule.
- Personalized fitness goals, progress tracking, and achievable challenges that encourage consistent physical activity.
- An engaging platform that combines financial competition with physical activity, allowing him to pursue both interests in one place.

Frustrations

Connor finds it difficult to balance his academic workload, career ambitions, and personal interests while maintaining an active lifestyle. Beginner-focused

financial platforms may not provide enough depth or challenge to keep him engaged, while traditional fitness applications may not offer the competitive elements that motivate him. Connor wants an efficient, challenging experience that allows him to demonstrate his financial knowledge while encouraging him to make time for exercise.

2.5.3 Daniella — The All-Arounder

Background

Daniella is a recent Civil Engineering graduate from the University of Guelph who gained valuable work experience and earned a substantial amount of money through multiple co-op placements throughout his degree. Through personal curiosity and working with her bank to understand her finances, Daniella developed a strong awareness of saving and managing money. Following graduation, she dedicated significant time to building her own investment portfolio and developing her financial knowledge. Although she has gained practical experience managing her savings, Daniella wants to become more confident in making independent investment decisions.

Goals and Motivations

Daniella enjoys staying active through a variety of organized sports which he participates in throughout the year. She worries that the time commitment required by multiple sports leagues may eventually become unsustainable, and makes the decision to cut back on some of the organized leagues that she participates in. She is interested in beginning a more traditional circuit-style workout as these tend to be more convenient. Financially, she wants to continue growing her investment portfolio while gaining more experience evaluating opportunities and making important investment decisions independently.

Behaviours and Habits

Daniella balances her professional responsibilities and personal interests while maintaining an active social life through organized sports. She prefers physical activities that are engaging, competitive, and social. Financially, Daniella takes an active role in monitoring his savings and managing his investment portfolio. She is curious, self-motivated, and willing to learn independently,

but recognizes that managing a portfolio with real money offers limited opportunities to experiment with unfamiliar strategies without risking her savings.

Needs and Expectations

- A realistic stock trading simulator that allows her to practice making independent investment decisions without risking real money.
- Investment tools that help evaluate market trends, compare opportunities, and understand the potential risks and outcomes of different strategies.
- Performance tracking and feedback to evaluate investment decisions and identify areas for improvement.
- Convenient fitness tracking and personalized activity goals that fit her professional responsibilities and sports league commitments.
- Flexible exercise recommendations that provide alternatives to traditional gym workouts and help maintain fitness during busy periods.
- An integrated platform that encourages both financial development and physical activity without requiring excessive time or effort.

Frustrations

Daniella finds it challenging to balance his professional life, multiple sports leagues, and financial goals without over-committing her time. Although she enjoys sports, she recognizes that relying entirely on organized leagues may make it difficult to maintain a consistent fitness routine year-round. Similarly, while she has successfully developed her own investment portfolio, she wants more opportunities to test strategies and learn from her decisions without putting his savings at risk. Daniella wants a flexible and practical solution that supports her financial growth and physical well-being while fitting naturally into his lifestyle.

Insert your content here.

2.6 Priorities Assigned to Users

The project's main priority will be to focus on the beginner - novice users of the application. The goal will be to teach the basis of health/fitness and financial literacy. *Insert your content here.*

2.7 User Participation

User participation will be incorporated into the development process. User feedback will help to make design decisions and provide a better scope of features.

- User interviews planned for Term 1 (CD-Fall document) to help elicit requirements and plan for the development.
- User testing on demo builds planned for Term 2 (Weeks 17 and 21) to receive feedback on decisions and provide an outside source of feedback.

Insert your content here.

2.8 Maintenance Users and Service Technicians

Insert your content here.

3 Mandated Constraints

3.1 Solution Constraints

The constraints below limit how the product can be built. They come from decisions recorded in the Problem Statement and the Development Plan, and the design must satisfy them.

SC1: Android platform

Description: The product shall run as a native mobile application on Android phones.

Rationale: The team narrowed the scope from multiple operating systems to Android only so the product can be finished to a high standard within the capstone timeline. Support for iOS and other platforms is a stretch goal.

Fit Criterion: The application installs and runs all core features on an Android phone.

SC2: Frontend technology

Description: The mobile client shall be written in TypeScript using the React Native framework.

Rationale: The team agreed on this stack in the Development Plan, based on the members' existing experience. React Native also leaves room for an iOS build later without rewriting the client.

Fit Criterion: All client source code is TypeScript built on React Native.

SC3: Backend technology

Description: The backend API shall be built in Java with the Spring framework, and persistent data shall be stored in a MySQL relational database.

Rationale: Chosen in the Development Plan. A relational database keeps user, transaction, and portfolio records structured and quick to query.

Fit Criterion: The API is a Spring application and all persistent application data lives in a MySQL database.

SC4: Fitness data sources

Description: Activity data shall be read automatically from the phone's built-in health data. Data from external devices such as fitness watches or heart rate monitors shall only be read through an SDK or API published by the device's manufacturer.

Rationale: Automatic tracking avoids manual entry, which users could inflate to earn currency. Using official manufacturer interfaces keeps the integration supported and within the manufacturer's terms of use.

Fit Criterion: The product has no feature for manually entering step counts, and every external device integration uses a documented SDK or API from that device's manufacturer.

SC5: Simulated market only

Description: All trading shall take place in a simulated market using virtual currency. The product shall not handle real money, connect to a brokerage, or let users convert virtual currency into cash.

Rationale: The goal is to teach how investments grow, not to run a trading platform. Handling real money or real trades would bring financial regulation that is outside the scope of a capstone project.

Fit Criterion: No feature of the product accepts, stores, or pays out real money, and no trade is sent to a real exchange or broker.

SC6: Distribution

Description: The product shall be distributed to users through the Android application store.

Rationale: The Problem Statement specifies that distribution is handled by the device's application store, which also handles installs and updates.

Fit Criterion: The release build meets the store's publishing requirements and can be installed from it.

3.2 Implementation Environment of the Current System

Flexin is a new product and does not replace an existing system. Today, people who want to build both habits use separate tools: a fitness tracker or the phone's own health app for activity, and a stock market simulator or a real brokerage for investing. Nothing links the two.

The product will run on the environment its users already have:

- An Android phone with Health Connect, which collects activity data from the phone's sensors and from other fitness apps the user has installed.
- Optionally, a wearable such as a Garmin watch, or an account with a fitness service such as Strava.

- A mobile data or Wi-Fi connection to reach the Flexin backend.

The backend (Spring API and MySQL database, see SC3) will be hosted on servers run by the team.

3.3 Partner or Collaborative Applications

The product depends on the following external applications for fitness data. Each integration must follow the provider's developer terms and permission model (see SC4).

Health Connect (Android): The main source of step counts and workout data. The user must grant Flexin permission to read each data type.

Garmin Connect IQ: Gives access to activity data from Garmin watches for users who own one.

Strava API: Gives access to runs, rides, and other activities logged in Strava, after the user authorizes Flexin on their Strava account.

The product also needs a source of asset prices for the simulated market. The team will choose it at the Market Decision milestone in Week 6 (see Section 3.6). Whatever source is chosen must allow use in an educational application at no cost to the team.

3.4 Off-the-Shelf Software

The product will be built on the following existing software rather than writing the equivalent from scratch:

- React Native and TypeScript for the mobile client.
- Java Spring for the backend API.
- MySQL for the database.
- JUnit for backend unit and system tests, and Detox for end-to-end tests of the mobile client.
- ESLint and the linters built into the team's IDEs.

- Figma for UI mockups.
- GitHub for source control, issue tracking, and CI/CD.

Any library added to the product must have a license that allows it to be included in proprietary software, since the project uses a proprietary license (see Section 3.8).

3.5 Anticipated Workplace Environment

Flexin is used on the user's phone, wherever they are. That affects the design in a few ways:

- Users will often check the app in short sessions, for example after a walk or workout, while outdoors, or while moving. Key screens need to be readable on a small screen in bright light and usable with one hand.
- The connection may be slow or drop out. Activity data must not be lost when the phone is offline; it should sync once the connection returns.
- Users will not have a manual or training. The interface has to explain financial terms itself, in line with the goal of a simple and delightful user experience.
- Step counting runs in the background through Health Connect, so the app must not drain the battery noticeably during a normal day.

3.6 Schedule Constraints

The project runs on the capstone course schedule, from September 2026 to April 2027. Dates cannot move. The milestones that constrain development are:

Week of	Milestone
Oct 19, 2026	SRS and Hazard Analysis Rev 0; preliminary market engine working for the proof of concept
Nov 9, 2026	Virtual currency logic defined (how activity converts into currency)
Nov 23 – Dec 4, 2026	Proof of concept demo
Jan 18, 2027	Design Document Rev 0
Jan 25, 2027	Demo build ready for first round of user testing
Feb 1 – Feb 12, 2027	Rev 0 demo
Mar 1, 2027	Second round of user testing
Mar 15, 2027	Final demo
Mar 29, 2027	Final documentation (Rev 1)
Apr 5, 2027	Capstone EXPO

Features that cannot be finished by the Rev 0 demo with enough time left for user testing will be moved to the stretch goals rather than delaying the schedule.

3.7 Budget Constraints

The project has no dedicated budget. All software, services, and hosting must be free, or covered by free tiers, student plans, or resources provided by the course. This rules out paid market data feeds and paid hosting, and any integration whose API charges per request beyond a free tier.

Development and testing will use the team members' own Android phones and the Android emulator, so no test hardware will be bought.

3.8 Enterprise Constraints

- **Course requirements.** The project must meet the deliverables and standards of the McMaster capstone course, including the document templates and checklists provided by the instructor.
- **Licensing.** The project uses a proprietary license that reserves all rights to the team (see the `LICENSE` file). This limits which third-party code can be included (see Off-the-Shelf Software).

- **Personal data.** The product stores personal and health data, such as height, weight, health conditions, and activity history. It must handle that data in line with Canadian privacy law (PIPEDA): collect only what is needed, get the user's consent, and let users see and delete their data. Details will be set out in the security and privacy requirements.
- **No financial advice.** The product teaches general investing concepts. It must not present anything as personalized financial advice, and any content that could be read that way needs a clear disclaimer.
- **App store policies.** The product must meet Google Play's developer policies, including its rules for health data and for finance-related apps.

4 Naming Conventions and Terminology

4.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

Insert your content here.

5 Relevant Facts And Assumptions

5.1 Relevant Facts

Insert your content here.

5.2 Business Rules

Insert your content here.

5.3 Assumptions

Insert your content here.

6 The Scope of the Work

6.1 The Current Situation

The modern software landscape treats physical health management and financial education as separate domains, despite both relying on consistent daily actions and the long-term benefits of compounding. Existing fitness applications, such as Strava, Garmin Connect, and Google Fit, track physical activity through metrics such as steps, heart rate, and GPS routes, while using badges, streaks, and milestones to encourage engagement. However, the long-term effectiveness of these incentives varies, and meaningful health benefits often require sustained participation over time **??**.

Similarly, financial simulators provide users with risk-free investment experiences but may rely on virtual capital that is disconnected from real-world effort and financial constraints. These environments can allow users to experiment with investing without experiencing the consequences of actual financial decisions. Furthermore, their educational value depends on users' motivation to learn, rather than providing a direct incentive to develop stronger personal financial management habits.

Financial literacy remains a significant challenge in Canada. Boisclair et al. found that only 42% of Canadian respondents correctly answered three foundational questions concerning compound interest, inflation, and risk diversification **?**. The same study found that respondents who answered all three questions correctly were 10 percentage points more likely to have retirement savings, demonstrating an association between financial knowledge and retirement preparedness **?**.

Despite their shared emphasis on consistency and compounding, physical activity tracking and financial education remain largely disconnected. Fitness data is primarily managed through health-focused platforms, while investment simulators do not typically connect simulated capital to users' real-world effort or habit formation. Consequently, users lack an integrated way to explore how small, repeated actions can contribute to long-term outcomes across both domains.

This gap presents an opportunity for an integrated platform that connects physical activity with simulated investing. By linking consistent activity to incremental virtual investment opportunities, such a platform could introduce more tangible short-term incentives, demonstrate the effects of compounding, and teach financial concepts through familiar daily habits. This

approach could encourage sustained participation while helping users develop the financial literacy needed for informed long-term decision-making.

6.2 The Context of the Work

The Flexin system connects physical activity tracking with simulated investing by converting validated exercise data into virtual investment capital. Users can use this capital to build simulated portfolios containing fictional stocks, exchange-traded funds (ETFs), funds, bonds, and other supported assets without risking real money. Flexin integrates with Android Health Connect and Garmin to retrieve supported fitness data, while its internally developed financial market engine generates fictional assets, prices, and market events. The system also provides portfolio management, financial education, and notifications.

6.2.1 Project Context Diagram

Figure 1 illustrates the interactions between Flexin and its users, fitness integrations, and device environment. The custom financial market engine is an internal component of Flexin that generates simulated financial data rather than retrieving real-world market prices.

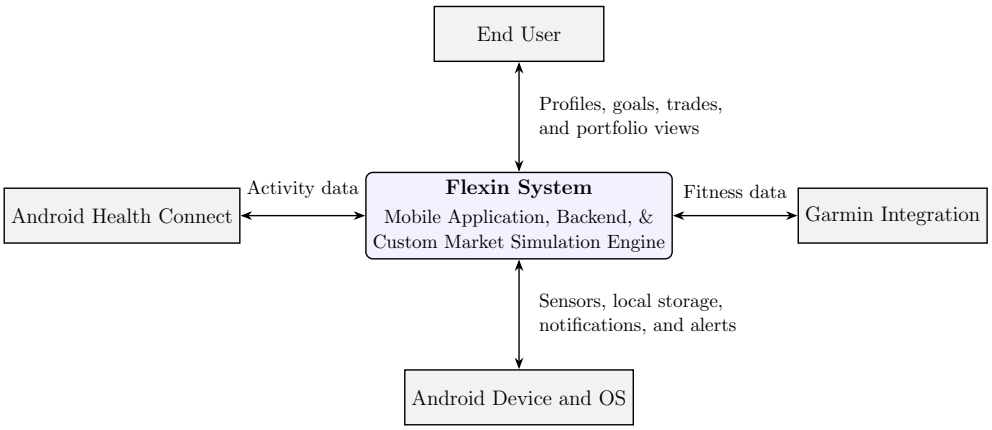


Figure 1: Flexin Project Context Diagram

6.2.2 Adjacent Systems and External Entities

The Flexin system interacts with the following entities and services.

Entity	**Responsibility**
End User	Provides profile information, exercise goals, and simulated trade instructions; receives portfolio insights, educational guidance, and virtual currency balances.
Android Health Connect	Provides aggregated health and activity data, such as step counts and exercise metrics, subject to device compatibility and user permissions.
Garmin Integration	Retrieves supported fitness and workout data from compatible Garmin devices and services, subject to API access and user authorization.
Custom Simulated Financial Market Engine	An internally developed component that creates and manages fictional stocks, ETFs, funds, bonds, and other supported assets. It generates simulated prices, market events, and asset performance without relying on real-world market data.
Android Device and OS	Provides the runtime environment, network connectivity, local storage, and native notification services.

Table 1: Flexin Adjacent Systems and External Entities

6.2.3 Information Flows Across the Work Boundary

Inbound Information. Inbound information includes user commands, profile and configuration data, exercise goals, and simulated trade orders. Flexin also receives fitness telemetry, such as step counts and workout records, through Android Health Connect and the Garmin integration. The custom financial market engine supplies internally generated asset prices, historical price series, simulated market events, and distributions such as dividends or fund returns.

Outbound Information. Outbound information includes virtual currency earned through validated activity, portfolio balances, asset allocations, and investment performance visualizations. Flexin also provides contextual fi-

nancial education on compound growth, diversification, and investment risk. Optional leaderboards, activity milestones, workout reminders, and simulated market notifications encourage continued engagement.

6.2.4 Boundary Scope and Enclosure

In Scope. Flexin includes fitness data validation, activity-to-currency conversion, duplicate reward prevention, simulated portfolio management, and virtual trade execution. Its custom financial market engine generates fictional assets and determines their simulated prices and performance. The backend manages accounts, asset positions, transaction records, and portfolio valuations, while the React Native and TypeScript frontend presents fitness and investment dashboards. An educational rules engine provides contextual explanations of financial concepts.

Out of Scope. Flexin does not depend on real-world stock prices, execute actual financial transactions, or manage real investment funds. It does not control the accuracy or availability of third-party fitness data, Garmin services, Android Health Connect, or device hardware. External fitness integrations remain subject to provider capabilities, API availability, device compatibility, and user permissions.

6.3 Work Partitioning

Work partitioning divides the Flexin system into Business Use Cases (BUCs), each representing a business event that the system must handle. These use cases cover user onboarding, fitness data integration, virtual currency allocation, simulated investing, financial education, and user engagement. Table 2 summarizes the business events, their triggers, expected outcomes, and associated use cases.

Event	Business Event / Trigger	Result / Output	Associated BUC
BE-01	User registers and submits baseline goals	Account and virtual wallet initialized	BUC 1: User Onboarding
BE-02	Activity data arrives from Health Connect or Garmin	Validated activity log updated	BUC 2: Activity Ingestion
BE-03	User achieves an activity reward threshold	Virtual currency credited	BUC 3: Currency Allocation
BE-04	User submits a simulated buy order	Asset holdings and virtual balance updated	BUC 4: Asset Purchase
BE-05	User submits a simulated sell order	Holdings reduced and sale proceeds credited	BUC 5: Asset Sale
BE-06	Scheduled market simulation cycle begins	Fictional asset prices and valuations updated	BUC 6: Market Simulation
BE-07	User requests portfolio analytics	Portfolio growth and activity contributions displayed	BUC 7: Portfolio Analytics
BE-08	User requests financial guidance	Contextual educational content displayed	BUC 8: Financial Education
BE-09	User accesses community rankings	Opt-in leaderboard displayed	BUC 9: Community Leaderboard
BE-10	Reminder schedule or simulated market event is triggered	Relevant notification delivered	BUC 10: Notifications

Table 2: Flexin Business Events and Associated Use Cases

6.4 Specifying a Business Use Case (BUC)

This section specifies each Flexin Business Use Case (BUC), including its goal, actors, trigger, preconditions, main flow, alternative flows, and post-conditions.

BUC 1: User Onboarding

Goal: Create a user account and initialize their Flexin profile.

Actors: End User.

Trigger: The user opens Flexin and selects the registration option.

Preconditions: The application is available and the user does not already have an active account.

Main Flow:

1. The user submits the required registration details.
2. Flexin validates the details and creates the account.
3. The user provides any requested baseline goals or preferences.
4. Flexin initializes the user's profile and virtual wallet.

Alternative Flows: If required details are missing or invalid, Flexin displays an error and requests corrections. If an account already exists, the user is prompted to sign in.

Postconditions: The user account and initial virtual wallet are created successfully.

BUC 2: Activity Ingestion

Goal: Retrieve and validate fitness activity data from supported integrations.

Actors: End User, Android Health Connect, Garmin Integration.

Trigger: New activity data becomes available or the user initiates an activity data synchronization.

Preconditions: The user is registered and has granted the required permissions to an available supported integration.

Main Flow:

1. Flexin retrieves available activity data from Health Connect or Garmin.
2. Flexin validates the data and checks for duplicate activities.
3. Valid activities are stored in the user's activity history.
4. The application updates the user's activity summary.

Alternative Flows: If permissions are missing, the user is prompted to grant access. If an integration is unavailable or data is invalid, Flexin reports the issue without recording the invalid activity.

Postconditions: Valid activity data is stored without duplicate records.

BUC 3: Currency Allocation

Goal: Reward eligible physical activity with virtual currency.

Actors: End User, Flexin Reward System.

Trigger: A validated activity satisfies a configured reward threshold.

Preconditions: The user has an active account and the activity has been validated.

Main Flow:

1. Flexin evaluates the activity against the reward rules.
2. The system calculates the eligible virtual currency reward.
3. Flexin credits the user's virtual wallet and records the transaction.
4. The user can view the updated balance and reward details.

Alternative Flows: If the activity does not meet the reward threshold, no currency is awarded. If the activity has already been rewarded, Flexin prevents duplicate crediting.

Postconditions: The wallet balance and transaction ledger reflect any eligible reward.

BUC 4: Asset Purchase

Goal: Allow the user to purchase fictional financial assets using virtual currency.

Actors: End User, Flexin Market Engine.

Trigger: The user submits a simulated buy order.

Preconditions: The user is registered, the selected asset exists, and a valid simulated price is available.

Main Flow:

1. The user selects an asset and enters the purchase quantity.
2. Flexin calculates the simulated purchase cost.

3. The system verifies that the user has sufficient virtual currency.

4. Flexin executes the simulated order and updates the wallet and holdings.

5. The completed transaction is recorded and displayed to the user.

Alternative Flows: If the balance is insufficient, the order is rejected. If the asset or price is unavailable, the user is informed and the transaction is not executed.

Postconditions: For a successful purchase, the virtual balance is reduced and the user's asset holdings increase accordingly.

BUC 5: Asset Sale

Goal: Allow the user to sell fictional assets and receive virtual currency.

Actors: End User, Flexin Market Engine.

Trigger: The user submits a simulated sell order.

Preconditions: The user holds a sufficient quantity of the selected asset and a valid simulated price is available.

Main Flow:

1. The user selects an asset and enters the quantity to sell.

2. Flexin calculates the simulated sale proceeds.

3. The system verifies that the user holds sufficient units.

4. Flexin executes the simulated order and updates the holdings and wallet.

5. The completed transaction is recorded and displayed to the user.

Alternative Flows: If the user holds insufficient units or the asset price is unavailable, the order is rejected without changing the portfolio.

Postconditions: For a successful sale, the user's holdings decrease and the corresponding virtual currency is credited.

BUC 6: Market Simulation

Goal: Generate and update fictional asset prices and market events.

Actors: Flexin Market Engine, Flexin Scheduler.

Trigger: A scheduled market simulation cycle begins.

Preconditions: The market engine is operational and fictional assets have been initialized.

Main Flow:

1. The scheduler initiates a simulation cycle.
2. The engine generates price changes and fictional market events according to the system's simulation rules.
3. Flexin updates asset valuations and relevant market information.
4. Updated prices and events become available to portfolio and trading functions.

Alternative Flows: If a simulation cycle fails, Flexin records the failure and retains the last valid market state until an update succeeds.

Postconditions: The latest valid fictional asset prices and market events are available within Flexin. The engine does not rely on real-world financial market data.

BUC 7: Portfolio Analytics

Goal: Present the user with portfolio performance and activity-related progress.

Actors: End User.

Trigger: The user opens the portfolio analytics view.

Preconditions: The user is registered and portfolio or activity data is available.

Main Flow:

1. Flexin retrieves the user's virtual balance, holdings, and transaction history.
2. The system calculates portfolio value and performance metrics using simulated asset prices.
3. Flexin retrieves relevant activity and reward information.

4. The application displays the resulting portfolio and activity summaries.

Alternative Flows: If insufficient history exists, Flexin displays available information and indicates that more data is needed for certain metrics.

Postconditions: The user can view their latest available portfolio and activity analytics.

BUC 8: Financial Education

Goal: Provide educational guidance relevant to the user's simulated investing activity.

Actors: End User, Flexin Educational Rules Engine.

Trigger: The user requests financial guidance or opens an educational feature.

Preconditions: The user can access the application.

Main Flow:

1. Flexin identifies the context of the user's request.
2. The educational rules engine selects relevant educational content.
3. Flexin presents explanations or guidance to the user.
4. The user reviews the content and may return to the application.

Alternative Flows: If no specific guidance matches the request, Flexin presents general educational content.

Postconditions: Relevant educational content is displayed. Guidance is educational and does not execute trades on the user's behalf.

BUC 9: Community Leaderboard

Goal: Allow users to view community rankings based on eligible activity or simulated portfolio metrics.

Actors: End User.

Trigger: The user opens the community leaderboard.

Preconditions: The user has access to Flexin and has enabled any participation required for their own information to appear in rankings.

Main Flow:

1. Flexin retrieves eligible ranking data from participating users.

2. The system calculates or retrieves rankings using the configured metrics.

3. The application displays the leaderboard and its ranking criteria.

Alternative Flows: If the user has opted out or no ranking data is available, Flexin displays the available leaderboard information without including the user's opted-out data.

Postconditions: The user can view the available community rankings in accordance with participation and privacy settings.

BUC 10: Notifications

Goal: Notify users about relevant activity rewards, reminders, and simulated market events.

Actors: End User, Flexin Notification Service, Android Device and OS.

Trigger: A configured reminder, eligible reward, or relevant simulated market event occurs.

Preconditions: The relevant event or schedule exists, and the user has enabled the applicable notifications and granted any required device permissions.

Main Flow:

1. Flexin identifies the event and determines whether a notification should be sent.

2. The system generates a notification containing relevant information.

3. Flexin sends the notification through the supported device mechanism.

4. The user receives the notification if device settings and permissions permit delivery.

Alternative Flows: If notifications are disabled or delivery fails, Flexin does not guarantee delivery and may retain the event for in-app display.

Postconditions: The notification is submitted for delivery or the delivery failure is handled according to the system's notification rules.

7 Business Data Model and Data Dictionary

This section outlines the business data model and data dictionary for the project. The business data model provides a high-level overview of the data relevant to the business, while the data dictionary specifies the content and structure of the data elements within the system (i.e. classes, attributes, relationships). Together, these components serve as a foundation for understanding how data is organized, stored, and utilized within the business processes.

7.1 Business Data Model

This business data model is a high-level diagram of the data that is relevant to the business. It identifies the key entities and the relationships between them. The model serves as a blueprint for understanding how data flows through the business processes and how it is stored and managed. Due to still being in the earlier stages of the project, this model may be subject to changes in the later stages of the project. This business data model is not a complete representation of the data, but rather a simplified view that highlights the most important aspects of the business data. It is intended to provide a clear and concise overview of the data landscape for stakeholders and team members.

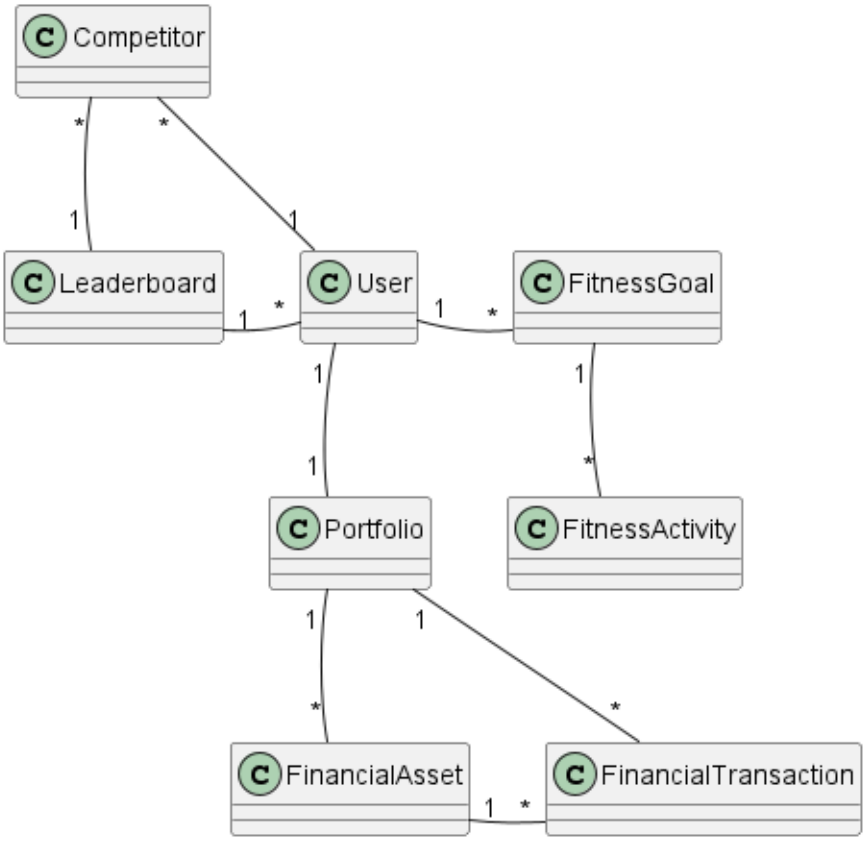


Figure 2: Business Data Model

Each of the rectangles represents a class of business data. The attributes of each class are defined in the data dictionary.

7.2 Data Dictionary

7.2.1 User

Description: Represents a registered user of the system.

Attribute	Type	Description	Constraints
User ID	Integer	Unique identifier for a user account	Primary Key, Not Null

Password	String	User authentication credential	Encrypted, Not Null
Currency	Decimal	Current virtual currency balance	≥ 0
Portfolio ID	Integer	Associated portfolio identifier	Foreign Key
Experience Points	Integer	Accumulated experience points earned through platform activities	≥ 0
Active Score	Integer	Current activity score used for engagement and competition metrics	≥ 0

7.2.2 Competitor

Description: Represents a user's competitors that may appear on leaderboards and challenges.

Attribute	**Type**	**Description**	**Constraints**
Competitor ID	Integer	Unique competitor record identifier	Primary Key
User ID	Integer	Associated user account	Foreign Key
Friend Flag	Boolean	Indicates whether the competitor is on the user's friends list	True/False

7.2.3 Leaderboard

Description: Stores ranking and competitive information for users.

Attribute	**Type**	**Description**	**Constraints**
Leaderboard ID	Integer	Unique leaderboard identifier	Primary Key

User ID	Integer	Associated user on the identified leaderboard	Foreign Key
User Score	Integer	User's competition score on the identified leaderboard	≥ 0
Leaderboard Rank	String	Tier ranking (e.g., Bronze, Silver, Gold)	Enumerated Values
User Rank	Integer	Position within the leaderboard	≥ 1

7.2.4 Fitness Goal

Description: Represents an individual fitness objective created by a user.

Attribute	**Type**	**Description**	**Constraints**
Goal ID	Integer	Unique goal identifier	Primary Key
User ID	Integer	Identifies which user has this goal	Foreign Key
Status	Enum	Current goal state	Active, Completed, Cancelled
Reward On Completion	Decimal	Virtual currency awarded upon completion	≥ 0

7.2.5 Fitness Activity

Description: Represents a fitness activity associated with achieving a fitness goal.

Attribute	**Type**	**Description**	**Constraints**
Activity ID	Integer	Unique activity identifier	Primary Key
Goal ID	Integer	Associated fitness goal	Foreign Key
Activity Name	String	Name of activity	Not Null

Activity Details	String	What is required/associated with carrying out the activity (i.e. 10,000 steps, burn 100 calories, etc.)	Not Null

7.2.6 Portfolio

Description: Represents a user's investment portfolio.

Attribute	**Type**	**Description**	**Constraints**
Portfolio ID	Integer	Unique portfolio identifier	Primary Key
Gains/Losses	Decimal	Current unrealized portfolio gain or loss	None
Asset ID	Integer	Financial asset within the identified portfolio	Foreign Key
Quantity	Integer	Quantity of asset held	≥ 0

7.2.7 Financial Asset

Description: Represents a simulated financial instrument available for trading.

Attribute	**Type**	**Description**	**Constraints**
Asset ID	Integer	Unique asset identifier	Primary Key
Cost To Buy/Sell	Decimal	Current purchase or sale price of the asset	≥ 0
Quantity Available	Integer	Total quantity available in the simulated market	≥ 0

7.2.8 Financial Transaction

Description: Records the purchase and sale of financial assets within a user's portfolio.

Attribute	Type	Description	Constraints
Transaction ID	Integer	Unique transaction identifier	Primary Key
Portfolio ID	Integer	Associated portfolio where the transaction occured	Foreign Key
Asset ID	Integer	Asset involved in the transaction	Foreign Key
Transaction Cost	Decimal	Cost of the asset at the time of transaction	≥ 0
Transaction Date	DateTime	Timestamp of transaction execution	Not Null

8 The Scope of the Product

This section will explore the different features of the application. This section explores the full scope of Flexin' application and will help to define main user functions.

8.1 Product Boundary

App Functionality

The Flexin application is designed as an educational tool to help young adults improve their personal fitness and financial literacy. The app provides a range of interactive features that allow users to engage with both domains in a meaningful and motivating way.

- Create a personal profile within the application. This profile allows users to:
    1. Enter personal information to tailor and control fitness goals.
    2. Select the desired level of difficulty and pace for financial learning modules.
- Enter, manage, and track fitness goals.

- Complete fitness challenges to earn virtual currency.
- Use virtual currency to invest in a simulated stock market environment to practice financial principles.
- View other users' progress and compete to build the largest virtual investment portfolio.

Excluded Functionality / Outside Scope

The Flexin' application will be developed specifically for the Android operating system and optimized for use within Canada. Support for other operating systems or international deployment is outside the scope of this project.

External Systems

To accurately track fitness goals, the application must retrieve data such as steps taken, heart rate, calories burned, and distance traveled. Modern Android devices provide these metrics natively, and the app will integrate with the Android Health Connect API to access this information. Additionally, many external fitness devices (e.g., Garmin watches) offer APIs and SDKs that provide detailed activity data. The *Flexin* app will communicate with these external systems to retrieve and incorporate user fitness data where supported.

8.2 Product Use Case Table

Insert your content here.

8.3 Individual Product Use Cases (PUC's)

Insert your content here.

9 Functional Requirements

9.1 Functional Requirements

Insert your content here.

10 Look and Feel Requirements

10.1 Appearance Requirements

ID	Requirement	Specification
LF-APP-01	Visual Hierarchy	**Description:** The dashboard shall present fitness metrics and financial metrics in a balanced, uncluttered layout using unified summary cards. **Rationale:** Equal emphasis reinforces the relationship between physical activity and simulated investment growth. **Fit Criterion:** All primary screens shall follow the Figma design library and its 4-point/8-point spacing grid.
LF-APP-02	Compounding Chart	**Description:** The portfolio chart shall distinguish cumulative workout earnings from simulated investment gains and dividends. **Rationale:** Users should understand the difference between contributions and compounded returns. **Fit Criterion:** Two visually distinct chart traces shall allow users to distinguish these values within 5 seconds.
LF-APP-03	Responsive Layout	**Description:** The interface shall adapt to Android screen sizes, aspect ratios, pixel densities, and system font scaling. **Rationale:** Flexin must remain usable across supported Android devices. **Fit Criterion:** On Android 10+ devices with 5.5–6.8-inch screens, no text shall clip or touch targets overlap, including at 130% font scaling.

ID	Requirement	Specification
LF-APP-04	Theme and Contrast	**Description:** The interface shall support Light and Dark modes with distinguishable fitness and financial status indicators. **Rationale:** Consistent contrast improves readability and accessibility in different environments. **Fit Criterion:** The interface shall provide sufficient colour contrast for readability, meeting WCAG 2.1 AA minimum ratios of 4.5:1 for normal text and 3:1 for large text and essential graphical elements.
LF-APP-05	Navigation	**Description:** A persistent bottom navigation bar shall link to Dashboard, Activity, Market/Portfolio, Learn, and Community. **Rationale:** Simple navigation reduces user effort. **Fit Criterion:** Primary operations shall be reachable within three taps from the home dashboard.

Table 11: Flexin Appearance Requirements

10.2 Style Requirements

ID	Requirement	Specification
LF-STY-01	Approachable Tone	**Description:** Typography, interface styling, and copy shall be welcoming and accessible. Financial terminology shall use clear language with contextual tooltips. **Rationale:** Reducing intimidation encourages financial learning and exploration. **Fit Criterion:** At least 80% of target-demographic evaluators shall rate the interface as approachable, supportive, or inviting.

ID	Requirement	Specification
LF-STY-02	Simulated Finance Clarity	**Description:** Financial screens, holdings, orders, and balances shall clearly identify all financial activity as simulated. **Rationale:** Users must not confuse Flexin with a real brokerage or financial account. **Fit Criterion:** Every screen displaying financial values shall show the virtual currency symbol (e.g., FLX) and a clearly visible "Simulated Market" label.
LF-STY-03	Non-Intrusive Gamification	**Description:** Milestone rewards shall use subtle, encouraging animations without disrupting normal use. **Rationale:** Rewards should support sustainable habits rather than distract users. **Fit Criterion:** Reward animations shall last no longer than 1.5 seconds and be immediately dismissible by tapping the background or swiping.
LF-STY-04	Component Consistency	**Description:** Buttons, forms, modals, tooltips, cards, and toggles shall follow a shared styling system. **Rationale:** Consistency builds trust and reduces the learning curve. **Fit Criterion:** All components shall conform to the master React Native theme, verified through linting and pull-request UI reviews.

Table 12: Flexin Style Requirements

11 Usability and Humanity Requirements

This section outlines the usability and humanity requirements for the system, including ease of use, personalization and internationalization, learning, understandability and politeness, and accessibility.

11.1 Ease of Use Requirements

ID	Requirement	Specification
UHR-EOU-01	Onboarding Completion Time	**Description:** The system shall enable new users to complete initial account setup and onboarding within 5 minutes. **Rationale:** Users should be able to begin using the application almost immediately without excessive setup requirements. Within 5 minutes allows users to maintain their engagement with the system. If the setup and onboarding take too long, users may become frustrated and disengaged. **Fit Criterion:** A first-time user shall be able to setup their account, complete onboarding and reach the main dashboard/screen within 5 minutes. This will be tested later in the project through user testing and observation.
UHR-EOU-02	Dashboard Accessibility	**Description:** The system shall provide direct access to fitness progress, portfolio information, and rewards through the main dashboard. **Rationale:** Users should not need to navigate through multiple screens to access frequently used information. **Fit Criterion:** Important high-level KPIs pertaining to fitness progress, portfolio value, and virtual currency balance shall be visible from the application's primary screen after login. Users should be able to answer questions like "How much currency do I have available?", "What percentage of my fitness goal is complete?" and "How is my portfolio performing?" within 1 minute after logging into the system.

ID	Requirement	Specification
UHR-EOU-03	Minimal Screen Navigation	**Description:** The system shall minimize navigation complexity by ensuring that portfolio management and lower-level metric viewing can be reached from the main dashboard within two taps. **Rationale:** Users should not need to navigate through multiple screens to access detailed metrics and adjust their portfolios. Excessive navigation increases user frustration and decreases engagement. **Fit Criterion:** Users shall be able to access detailed fitness statistics, portfolio management, virtual currency transactions, and goal tracking within two taps from the main dashboard.
UHR-EOU-04	Error Feedback and Recovery	**Description:** The system shall provide clear, actionable, and immediate feedback to the user whenever user actions cannot be completed. **Rationale:** Clear feedback helps users understand how to resolve issues and continue using the application. **Fit Criterion:** Whenever an error occurs (i.e. the user is prevented to complete an action), the user will be notified immediately with a user-facing error message that shall explain the problem in great detail and provide guidance for corrective action.

ID	Requirement	Specification
UHR-EOU-05	Positive Feedback	**Description:** The system shall provide immediate visual feedback following user actions to confirm whether the requested operation was completed successfully. **Rationale:** Users should be able to easily determine whether an action has been executed correctly without needing to manually verify the system state. Clear feedback increases user confidence and reduces confusion. **Fit Criterion:** All user-initiated actions, including goal updates, portfolio transactions, account modifications, and fitness device connections, shall display a confirmation message, notification, or visual state change within 2 seconds of successful completion.

ID	Requirement	Specification
UHR-EOU-06	Simple and Reliable Device Connectivity	**Description:** The system shall simplify the process of connecting supported fitness devices and applications and maintain strong and reliable connections to these devices. **Rationale:** Integration with existing fitness technologies is a core project goal. Users should be able to link their existing devices without requiring technical expertise. Once connected, the system shall maintain a stable connection to ensure continuous data synchronization so that the user is not constantly required to re-establish the connection - as this will cause user frustration. **Fit Criterion:** Users shall be able to perform a one-time action to connect a supported fitness data source through a guided setup workflow. Once connected the first time, the system shall not sever this connection more than three times per year without explicit user intervention.
UHR-EOU-07	Consistent User Interface	**Description:** The system shall maintain consistent visual design, terminology, and navigation patterns across all screens. **Rationale:** Consistency improves user familiarity and reduces the effort required to learn new functionality. **Fit Criterion:** Similar actions shall use consistent icons, terminology, colors, and interaction patterns throughout the application.

ID	Requirement	Specification
UHR-EOU-08	Learnability	**Description:** The system shall require no more than 10 minutes for users to become proficient in performing core tasks and navigating the application. **Rationale:** Users should not require an excessive amount of time to learn and use the application effectively. Excessive time required to learn how to use the core functionalities of the application may lead to user frustration and disengagement. **Fit Criterion:** First-time users shall be able to make their first investment, set one fitness goal, and view fitness and portfolio performance metrics within 10 minutes from landing on the main screen after login.
UHR-EOU-09	Efficient Re-Learnability	**Description:** The system shall require no more than 5 minutes for former users to re-learn how to execute core tasks and navigating the application. **Rationale:** Even after time away from the system, users should be able to pick back up right where they left off. If a user has to spend too much time re-learning the application, it may lead to frustration and discourage users from re-engaging with the system. **Fit Criterion:** Former users with at least 1 month of consistent daily usage shall be able to re-learn how to manage at least one investment, manage at least one fitness goal, and view fitness and portfolio performance metrics within 5 minutes of returning to the system after 6+ months of inactivity.

11.2 Personalization and Internationalization Requirements

ID	Requirement	Specification
UHR-PER-01	Personalized Fitness Goals	**Description:** The system shall allow users to configure fitness goals based on their personal fitness level and preferences. **Rationale:** Users will possess different fitness capabilities and activity levels. Personalized goals encourage engagement while remaining achievable. **Fit Criterion:** Users shall be able to modify daily and weekly fitness goals that will take effect the next time period.
UHR-PER-02	Customizable Notifications	**Description:** The system shall allow users to configure which notifications they receive and how frequently they are delivered. **Rationale:** Users have varying preferences regarding reminders and activity notifications. **Fit Criterion:** Users shall be able to independently enable or disable workout reminders, portfolio updates, and achievement notifications.
UHR-PER-03	Customizable Privacy Settings	**Description:** The system shall allow users to control the visibility of their fitness achievements, portfolio statistics, and leaderboard participation. **Rationale:** Users have different privacy expectations and comfort levels regarding public sharing of personal information. **Fit Criterion:** Users shall be able to independently configure visibility settings for all community-facing features that will take immediate effect.

ID	Requirement	Specification
UHR-PER-04	Customizable Competition Groups	**Description:** The system shall allow users to customize the set of users with whom they share and compare fitness and portfolio progress. **Rationale:** Users may have different comfort levels regarding competition and sharing personal achievements. Allowing users to select their competition group creates a more personalized and engaging experience where friends can motivate each other. **Fit Criterion:** Users shall be able to choose between private, friends-only, and public, community-wide, leaderboard participation.
UHR-INT-01	Canadian Time Zone Support	**Description:** The system shall automatically detect and utilize the user's local Canadian time zone when calculating daily goals, rewards, and streaks. **Rationale:** Users may access the application from any Canadian province or territory. **Fit Criterion:** Daily statistics and streak calculations shall correspond to the user's local device time zone.

11.3 Learning Requirements

ID	Requirement	Specification
UHR-LRN-01	Minimal User Training	**Description:** The system shall enable first-time users to successfully use the application's core functionality without requiring formal training. **Rationale:** The application targets users with varying expertises and should be intuitive for a broad audience. **Fit Criterion:** At least 80% of first-time users shall be able to synchronize fitness data, earn virtual currency, and purchase an investment asset without external assistance.
UHR-LRN-02	Learning Curve of Financial Concepts	**Description:** The system shall present investment concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on financial investing. This means that all users, regardless of their prior financial experience, should be capable of learning and understanding the financial concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time investors shall successfully complete their first virtual investment and view their portfolio's growth after one week without assistance.

ID	Requirement	Specification
UHR-LRN-03	Learning Curve of Fitness Based Concepts	**Description:** The system shall present fitness concepts in a manner that can be understood by users with little or no prior experience. **Rationale:** One of the system's primary goals is to educate users on fitness and health related activities. This means that all users, regardless of their prior fitness experience, should be capable of learning and understanding the fitness concepts presented. **Fit Criterion:** During user testing, at least 80% of first-time participants in a new fitness activity shall successfully accomplish their set goal and view their health metrics after one week without assistance.
UHR-LRN-04	Integrated Educational Assistance	**Description:** The system shall provide contextual learning resources within the application. **Rationale:** In the event that users encounter unfamiliar fitness or financial terminology while using the platform, the system should offer immediate assistance to enhance their understanding so that users can quickly learn and pick up new concepts. **Fit Criterion:** Users shall be able to access explanatory information for all core financial and fitness concepts directly from the screen where the concept appears.

ID	Requirement	Specification
UHR-LRN-05	User Help Documentation	**Description:** The system shall provide readily accessible help documentation describing the application's functionality should any users require it. **Rationale:** Users should be able to resolve common questions without external support. **Fit Criterion:** A help section shall be accessible from the application's settings menu and include explanations of all major features. Support teams should receive zero inquiries about basic system functionality.

11.4 Understandability and Politeness Requirements

Insert your content here.

11.5 Accessibility Requirements

ID	Requirement	Specification
UHR-ACC-01	Screen Reader Compatibility	**Description:** The system shall support screen reader technologies for all primary application workflows. **Rationale:** Users with visual impairments may rely on screen readers to navigate the application. Users with visual impairments should not be excluded from benefitting from this application. **Fit Criterion:** All interactive interface elements shall have descriptive accessibility labels that can be read aloud.

ID	Requirement	Specification
UHR-ACC-02	Scalable Text Support	**Description:** The system shall support device-configured text scaling without loss of functionality. **Rationale:** Users with reduced vision may require larger text sizes to comfortably use the application. **Fit Criterion:** All application screens shall remain usable when the device text size is increased to the maximum supported accessibility setting.
UHR-ACC-03	Color Independent Information	**Description:** The system shall not rely solely on color to communicate information or system status. **Rationale:** Users with color vision deficiencies may be unable to distinguish information represented exclusively through color. **Fit Criterion:** All status indicators, icons, buttons, graphs, and notifications shall include a secondary visual indicator in addition to color.
UHR-ACC-04	Touch Target Accessibility	**Description:** The system shall provide interface controls that are sufficiently large for reliable touch interaction. **Rationale:** Users with limited dexterity or motor impairments may have difficulty selecting small interface elements. **Fit Criterion:** All touch-interactive elements shall meet the minimum touch target size recommended by Android accessibility guidelines.

ID	Requirement	Specification
UHR-ACC-05	Alternative Notification Methods	**Description:** The system shall provide both visual notifications, audio cues, and haptic feedback for important events and shall not rely exclusively on one. **Rationale:** Users who are deaf or hard of hearing may not perceive audio notifications. Users with visual impairments may not be able to perceive visual notifications. Providing alternative notification methods ensures that all users can stay informed of important events. **Fit Criterion:** All important user notifications shall be easily accesible to all users within the application interface.
UHR-ACC-06	Inclusive Activity Tracking	**Description:** The system shall not require a single type of physical activity as the sole method of earning rewards. **Rationale:** Users with disabilities, injuries, temporary medical conditions, or differing fitness abilities may be unable to participate in certain forms of exercise. **Fit Criterion:** The system shall support multiple reward-eligible activity types and allow users to select activities appropriate to their abilities.

12 Performance Requirements

Unless stated otherwise, the timing requirements in this section are measured on a mid-range Android phone (one released within the last four years) with a typical mobile data or Wi-Fi connection.

12.1 Speed and Latency Requirements

PR-SL1: App launch

Description: The app shall show the home screen within 3 seconds of being opened from a closed state.

Rationale: Users will mostly open the app for short check-ins, such as after a walk or workout. A slow start makes those check-ins less likely.

Fit Criterion: Across 20 cold starts on the reference phone, the home screen appears in 3 seconds or less in at least 19 of them.

PR-SL2: Response to input

Description: The app shall respond to a tap or navigation action within 1 second.

Rationale: Supports the goal of a simple and delightful user experience. Delays longer than about a second make an app feel broken.

Fit Criterion: At least 95% of measured screen transitions and button responses complete within 1 second.

PR-SL3: Trades

Description: After a user confirms a buy or sell order, the updated holdings and currency balance shall be shown within 2 seconds.

Rationale: Users need to see the result of a trade straight away to connect the action with its effect on their portfolio.

Fit Criterion: At least 95% of test trades show the updated portfolio within 2 seconds of confirmation.

PR-SL4: Activity sync

Description: When the app is opened, activity recorded since the last sync shall be read and turned into currency within 10 seconds.

Rationale: The reward has to feel tied to the activity that earned it. The whole point of the product is to make a distant payoff feel immediate.

Fit Criterion: In testing, the currency balance reflects all activity available in Health Connect within 10 seconds of opening the app.

12.2 Safety-Critical Requirements

Flexin does not control physical equipment, so none of its failures can directly injure someone. The Hazard Analysis does identify ways the product could encourage harm, and the requirements below address them. Each requirement lists the safety requirements and hazards from the Hazard Analysis that it traces to.

PR-SC1: Daily earning limits

Description: The amount of currency a user can earn from activity in one day shall be capped, and activity beyond a healthy daily volume shall earn currency at a reduced rate.

Rationale: Without these limits, the product would reward users for exercising past safe limits, or while injured, to earn more.

Fit Criterion: No amount of activity recorded in a single day earns more than the daily cap, and activity beyond the healthy daily volume earns less per unit than activity below it.

Hazard Analysis: SR8 (H4.1, H5.1)

PR-SC2: No penalties for missed days

Description: The product shall not take away currency or holdings when a user misses a goal or a day of activity. Users shall be able to freeze their streak for days missed because of illness, injury, rest, or a delayed sync.

Rationale: The Hazard Analysis lists shame over missed goals as a psychological hazard. Losing earned progress or a long streak would make that worse, could push users to exercise while injured or unwell, and makes users feel they have failed through no fault of their own.

Fit Criterion: No feature of the product reduces a user's balance or holdings because of inactivity, and a user can freeze their streak so that a missed day does not reset it.

Hazard Analysis: SR8 (H3.1, H4.1, H11.1)

PR-SC3: Simulated market is labelled

Description: Every screen that shows prices, holdings, or returns shall state that the market is simulated and that results do not predict real returns. Each of these screens shall link to an explanation of what the simulation does not model, such as fees, taxes, and real-world volatility.

Rationale: The Hazard Analysis lists financial loss from users applying strategies from the app to real markets. Users who treat simulated results as real-world truth may lose real money.

Fit Criterion: A review of every market and portfolio screen finds the notice visible on each one, and the explanation of the simulation's limits can be reached from each one.

Hazard Analysis: SR11 (H6.1, H7.1)

PR-SC4: Not financial advice

Description: The product shall state that its content is for education only and is not financial advice. The statement shall be shown during sign-up, where the user must acknowledge it before continuing, and shall stay available from the settings screen. Lessons, tips, and any other content that discusses investing shall not recommend specific real-world investments to a user.

Rationale: The Hazard Analysis lists financial loss from users acting on what they learn in the app. Flexin is not a licensed financial advisor, so users must understand that nothing in the app is personal advice about their real money.

Fit Criterion: A new account cannot be created without acknowledging the statement, the statement can be reached from settings, and a review of all educational content finds no recommendations to buy or sell specific real-world investments.

Hazard Analysis: SR11 (H6.1, H7.1)

PR-SC5: Exercise injury warning

Description: The product shall warn users that physical activity carries a risk of injury, that they should exercise within their own limits, and that they should consult a doctor before starting a new exercise routine if they have health concerns. The product's terms of use shall state that users exercise at their own risk and that the Flexin team is not responsible for any injury that happens while using the app. The user must acknowledge both during sign-up before continuing.

Rationale: The product rewards physical activity, so some users will exercise more because of it. The Hazard Analysis assumes users only attempt exercises they are able to do safely, and places injuries from exercise outside the product's control. Users need to be told this directly. The responsibility statement is a legal matter as well as a safety one, so it should also be listed under the Legal Requirements in Compliance Requirements, and its exact wording should be checked against the course's guidance before release.

Fit Criterion: A new account cannot be created without acknowledging the injury warning and the terms of use, and both can be reached from the settings screen at any time.

PR-SC6: Safe goal progression

Description: Goals that the product raises automatically shall increase gradually and include regular rest days. When a goal is missed several times in a row, the product shall suggest lowering it. Users shall be able to lower or override any goal themselves.

Rationale: Goals that rise too quickly push users to overexert themselves, while goals that are missed again and again lead to shame and users giving up on exercise.

Fit Criterion: Automatic goals never rise by more than the set maximum step, every goal plan includes regular rest days, a lower goal is suggested after the set number of misses in a row, and a user can lower any goal from the goal screen.

Hazard Analysis: SR6 (H4.1, H4.2)

PR-SC7: Supportive messages and notifications

Description: All goal-related messages and reminders shall use supportive language that does not shame or guilt the user. Users shall be able to choose how often they get notifications and set quiet hours.

Rationale: Guilt-based messages and too many notifications harm users' mental health and encourage compulsive use of the app.

Fit Criterion: Every goal-related message passes a team review for shaming or guilt-based language, and in testing no notifications are sent during the user's quiet hours or more often than the frequency they chose.

Hazard Analysis: SR7 (H4.1, H10.1)

PR-SC8: Leaderboards

Description: Users shall be able to opt out of group leaderboards. Leaderboards shall rank users by progress towards their own goals, not by absolute performance.

Rationale: Ranking users by absolute performance causes shame and anxiety for those who rank poorly.

Fit Criterion: Users who opt out do not appear on any leaderboard, and rankings are computed from each user's percentage of their own goal.

Hazard Analysis: SR9 (H4.1, H8.1)

12.3 Precision or Accuracy Requirements

PR-PA1: Exact money values

Description: Currency balances, prices, and trade amounts shall be stored and calculated as exact decimal values, not floating-point numbers, and shown to two decimal places.

Rationale: Floating-point rounding errors build up over many trades and would show users balances that do not add up.

Fit Criterion: After a scripted sequence of 1,000 trades, the stored balance matches the expected balance exactly.

PR-PA2: Consistent currency awards

Description: The currency awarded for a set of activity data shall follow the published conversion rules exactly, and the same data shall always earn the same amount.

Rationale: Users must be able to trust that effort is rewarded fairly, and the team must be able to check that it is.

Fit Criterion: For a set of test activity records, the awarded currency matches the amount calculated by hand from the conversion rules.

PR-PA3: No double counting

Description: Activity reported by more than one source, such as the phone and a watch both counting the same steps, shall be counted only once.

Rationale: Counting the same steps twice would over-reward users who own more devices, which the Hazard Analysis lists as a cause of poor learning.

Fit Criterion: When the same walk is recorded by two sources, the currency earned equals the amount for one recording.

PR-PA4: Portfolio value

Description: The displayed portfolio value shall equal the sum of each holding's quantity times its current simulated price, plus uninvested currency.

Rationale: The portfolio graph is the main tool for teaching compounding, so it has to be correct.

Fit Criterion: For test portfolios, the displayed value matches the value calculated by hand to the cent.

PR-PA5: Workout dates

Description: Each workout shall be dated by the time it happened on the source device, not the time it was synced. When workout data arrives late, the affected streaks and rewards shall be recalculated.

Rationale: Users who exercised on time should not lose their streak or currency because their device synced late or across a day boundary.

Fit Criterion: Workouts synced after midnight or from a different time zone are recorded on the day they happened, and the affected streaks and rewards are updated once the late data arrives.

Hazard Analysis: SR5 (H3.1, H11.1)

PR-PA6: Realistic market simulation

Description: The market simulation shall be calibrated against historical market data, so that its volatility and downturns are consistent with real markets.

Rationale: Returns that are unrealistically smooth or high would teach users a false understanding of risk and compounding.

Fit Criterion: Over a long simulated period, the average annual return and volatility are consistent with historical data for a broad market index, and downturns occur.

Hazard Analysis: SR10 (H6.1)

12.4 Robustness or Fault-Tolerance Requirements

PR-RFT1: Offline activity

Description: Activity recorded while the phone is offline, or while the app is closed, shall be credited the next time the app syncs, for gaps of up to 7 days.

Rationale: Users exercise in places without signal and will not open the app every day. Losing that activity would discourage them.

Fit Criterion: Activity recorded during a 7-day gap with no syncing is fully credited at the next sync.

PR-RFT2: Atomic writes

Description: Every database write with more than one step shall either complete fully or be rolled back. A trade shall update both the currency balance and the holdings, at the price the user confirmed, or not happen at all.

Rationale: A write that fails halfway leaves balances, portfolios, or progress wrong. For example, a failed trade could take a user's currency without giving them the asset, or the other way around.

Fit Criterion: Tests that simulate a failure at each step of a multi-step write leave the database unchanged. When the connection or server is cut during a test trade, the balance and holdings afterwards match either the state before the trade or the state after it, and completed trades execute at the confirmed price.

Hazard Analysis: SR12 (H7.2, H9.2)

PR-RFT3: Failing data sources

Description: If a fitness data source is unavailable or returns invalid data, the app shall keep working with its other sources and credit no currency for the invalid data. A failed workout sync shall be retried automatically, and if it still fails, the app shall tell the user which source has a problem and why.

Rationale: Outages at Garmin, Strava, or Health Connect are outside the team's control and should not make the whole app unusable. Users should find out about a failed sync from the app, not from a broken streak.

Fit Criterion: With one source disabled or returning malformed data in testing, the app stays usable, retries the sync, then shows a message naming that source and the cause, and awards nothing for the bad data.

Hazard Analysis: SR4 (H3.1)

PR-RFT4: Backend unavailable

Description: If the backend cannot be reached, the app shall show the last data it received, marked as possibly out of date, instead of crashing or showing empty screens.

Rationale: Users on poor connections should still be able to see their progress.

Fit Criterion: With the backend stopped, the app opens, shows the last known balance and portfolio, and displays an offline notice.

PR-RFT5: Account errors

Description: When account creation or sign-in fails, the app shall show a clear error message that states the cause.

Rationale: Users should know why they cannot get into their account, instead of being locked out with no explanation.

Fit Criterion: Every failed account creation or sign-in in testing shows an error message that states the cause.

Hazard Analysis: SR4 (H1.1)

12.5 Capacity Requirements

PR-C1: Users

Description: The system shall support at least 500 registered users, with at least 100 using the app at the same time, without breaking the speed requirements above.

Rationale: This covers the expected load during user testing, the final demo, and the Capstone EXPO, where many people may try the app at once.

Fit Criterion: A load test with 100 simulated concurrent users meets PR-SL2 and PR-SL3.

PR-C2: History

Description: The system shall store at least two years of activity and trade history for each user.

Rationale: Compounding only shows up over long periods, so users need to be able to look back over their history.

Fit Criterion: A test account loaded with two years of daily activity and weekly trades can display its full history, and its screens still meet PR-SL2.

12.6 Scalability or Extensibility Requirements

PR-SE1: Adding fitness sources

Description: A new fitness data source shall be addable without changing the currency conversion logic.

Rationale: The team expects to add device integrations over time, and the Problem Statement lists integration with existing fitness technologies as a goal.

Fit Criterion: Adding a new source only requires changes inside the integration module.

PR-SE2: Adding asset types

Description: New types of assets shall be addable to the simulated market without changing the trading logic.

Rationale: The product starts with stocks, bonds, and ETFs, and other asset types may be added later.

Fit Criterion: Adding a new asset type only requires new market data and configuration, not changes to the buy and sell code.

PR-SE3: More server capacity

Description: The backend API shall keep no per-user state between requests, so more instances can be run if the number of users grows.

Rationale: If the product continues after the capstone, as the business plan explores, it will need to serve more users than PR-C1.

Fit Criterion: Two API instances running against the same database give the same results as one.

PR-SE4: Other platforms

Description: Code that depends on Android-only features shall be kept separate from the rest of the client.

Rationale: Support for iOS is a stretch goal, and keeping Android-specific code separate keeps that option open.

Fit Criterion: All calls to Android-only APIs are in a clearly marked platform layer.

12.7 Longevity Requirements

PR-L1: Supported Android versions

Description: The app shall target the Android version that Google Play requires for new apps at the time of release, and keep working on Android versions released through the end of the capstone in April 2027.

Rationale: Google Play stops accepting apps that target old Android versions, and the app must stay usable for the final demo and the EXPO.

Fit Criterion: The release build meets Google Play's target version requirement and passes the test suite on the newest Android version available in April 2027.

PR-L2: Keeping user history

Description: Activity and portfolio history shall be kept for as long as the user's account exists, unless the user deletes it.

Rationale: The product teaches that habits pay off over years. Deleting old history would remove the evidence of that.

Fit Criterion: The system has no automatic deletion of history for active accounts.

13 Operational and Environmental Requirements

13.1 Expected Physical Environment

The application will be operable through mobile smartphones and smart tablets with cellular/Wi-Fi connectivity. Inputs will be completed through health tracking services that will be provided by the smart device that is running the software, or optionally received through other technologies used for health tracking.

13.2 Wider Environment Requirements

The app will be designed for the Android operating system for the Capstone Project. We must adhere to regulations/policies and limitations of the system. Software distribution and update management will be handled through the device specific application store (i.e. Google Play Store and various Android App Stores).

13.3 Requirements for Interfacing with Adjacent Systems

Many outside systems will be used:

- HTTPS connection to external APIs
    1. External Technology Connection (e.g., Garmin API, Strava SDK)
    2. Software-specific API for health-related tracking (Android Health Connect)
- HTTPS connection to internal APIs
    1. Java SpringBoot API for storing user data and retrieving from the database
- Database Connection
    1. Privately hosted MySQL server

13.4 Productization Requirements

ID	Requirement	Specification
PROD-01	Installation and Updates channels	**Description:** The application distribution and update management shall be handled through the supported Android app stores. **Rationale:** The app store system is an established process that users will already be familiar with. This allows the user base to continue a process they have become accustomed too. **Fit Criterion:** Users will be able to download and update the application through the Android app store.

ID	Requirement	Specification
PROD-02	Build and Quality Assurance	**Description:** The application will be subject to all the basic coding tests and standards needed to be published on the Android app store. **Rationale:** An app must be whole and completely working to provide value to the user base. **Fit Criterion:** All automated and manual written code tests must pass (100%) to be deemed fully complete.

13.5 Release Requirements

14 Maintainability and Support Requirements

This section outlines the requirements for maintaining and supporting the software product throughout its lifecycle. It includes specifications for maintenance procedures, support services, and any necessary adaptability procedures to ensure the product remains functional and up-to-date.

14.1 Maintenance Requirements

ID	Requirement	Specification
MAS-MNT-01	Fitness Activity Addition Effort	**Description:** The system shall support the addition of new reward-eligible fitness activity types with minimal maintenance effort. **Rationale:** Future releases may introduce additional forms of exercise and adaptive fitness activities. **Fit Criterion:** A new activity type shall be implemented, deployed, and fully tested within 2 months.

ID	Requirement	Specification
MAS-MNT-02	Investment Asset Addition Effort	**Description:** The system shall support the addition of new simulated investment assets with minimal maintenance effort. **Rationale:** Future versions of the application may expand the types of simulated investments available to users. **Fit Criterion:** A new investment asset shall be implemented, deployed, and fully tested within 2 months.
MAS-MNT-03	Corrective Maintenance Effort	**Description:** The system shall efficiently handle and correct software defects. **Rationale:** Timely defect resolution improves reliability and user satisfaction. **Fit Criterion:** Critical production defects shall be fixed and deployed within 5 business days of of the defect being reported.
MAS-MNT-04	Third-Party Integration Maintenance	**Description:** The system shall support updates to external fitness platform integrations with minimal impact on overall system functionality. **Rationale:** External APIs may change independently of the application and require maintenance. **Fit Criterion:** Changes required to maintain compatibility with a supported fitness provider shall be deployed within 5 business days of identifying the incompatibility.

ID	Requirement	Specification
MAS-MNT-05	Traceability	**Description:** The system shall provide sufficient documentation and traceability to allow future developers to efficiently understand previous changes and maintenance activities. **Rationale:** Accurate and up-to-date documentation improves maintainability by allowing future developers to understand previous modifications, known issues, and the reasoning behind architectural and implementation decisions. **Fit Criterion:** For every production release, developers shall be able to identify the implemented changes, associated issues, and documented design decisions within 30 minutes of searching project records.

14.2 Supportability Requirements

ID	Requirement	Specification
MAS-SUP-01	Backup and Recovery Support	**Description:** The system shall support recovery of user and application data following data corruption or infrastructure failure. **Rationale:** Effective recovery capabilities minimize the impact of data loss incidents on users. **Fit Criterion:** The system shall be capable of restoring application data from the most recent successful backup within 24 hours. Data backups must successfully occur at least once daily.

ID	Requirement	Specification
MAS-SUP-02	Self-Service User Support	**Description:** The system shall provide integrated help resources that allow users to resolve FAQs without requiring direct support from developers. **Rationale:** Users should be able to independently resolve common issues related to account management or recovery, activity tracking, portfolio management, and virtual currency rewards. **Fit Criterion:** Help documentation covering account management and recovery, activity synchronization, portfolio management, and reward allocation shall be accessible from within the application at all times. The user shall be able to successfully integrate these recovery protocols without developer intervention.
MAS-SUP-03	Contact Support Capability	**Description:** The system shall provide users with a mechanism to submit feedback or support requests for more complicated issues. **Rationale:** Some issues may require assistance beyond self-service documentation. **Fit Criterion:** Users shall be able to submit feedback or support requests from within the application at any time. These requests must be processed within 2 business days with a response sent back to the user regarding details pertaining to the plans and timeline to address the issue.

14.3 Adaptability Requirements

ID	Requirement	Specification
MAS-ADPT-01	External Fitness Provider Adaptability	**Description:** The system shall support the addition of new fitness data providers without requiring major changes to user workflows. **Rationale:** The ecosystem of fitness tracking applications and devices will evolve over time and additional providers may become desirable. **Fit Criterion:** Users shall be able to successfully connect newly supported fitness providers using the same workflow used for existing providers.
MAS-ADPT-02	Mobile Platform Portability	**Description:** The system shall support future deployment on additional mobile platforms beyond the initially targeted platform - Android devices. **Rationale:** Initial deployment is limited to Android, but future releases may target additional mobile operating systems, such as iOS. **Fit Criterion:** User account information and application data shall remain portable across supported platforms.
MAS-ADPT-03	Future AI Service Adaptability	**Description:** The system shall support future integration of AI-assisted educational content. **Rationale:** With the rapid growth of AI technologies, future releases may provide personalized fitness and financial learning experiences using AI technologies. **Fit Criterion:** AI-powered educational services shall be deployable without requiring modifications to existing user accounts.

15 Security Requirements

15.1 Access Requirements

ID	Requirement	Specification
SEC-ACC-01	Authentication and Sessions	**Description:** Users shall authenticate before accessing personal fitness data, portfolios, or transactions. The React Native client shall authenticate with the Spring API using secure token-based sessions. **Rationale:** Authentication protects personal data and virtual assets from unauthorized access. **Fit Criterion:** All protected API endpoints shall reject missing or invalid access tokens with HTTP 401. Access tokens shall expire within 60 minutes, with refresh or re-authentication required.
SEC-ACC-02	Role-Based Authorization	**Description:** Role-based access control shall restrict users to their own records and limit administrative or evaluator access to authorized functions. **Rationale:** Access restrictions prevent unauthorized modifications and cross-account data exposure. **Fit Criterion:** JUnit security tests shall confirm that attempts by standard users to access or modify another user's records return HTTP 403 in all tested cases. **Hazard Analysis:** SR3 (H2.1, H8.1, H9.1)

ID	Requirement	Specification
SEC-ACC-03	Fitness Integration Authorization	**Description:** Flexin shall request only necessary read permissions for supported integrations, Android Health Connect and Garmin. Integration credentials shall be stored using secure, Android Keystore-backed storage. **Rationale:** Secure credential handling protects external accounts and limits access to fitness data. **Fit Criterion:** Integration tokens shall use Android Keystore-backed secure storage. No credentials, tokens, or secrets shall be committed to the project repository.
SEC-ACC-04	Password and Sign-In Protection	**Description:** Flexin shall enforce a minimum password strength, offer optional multi-factor authentication through the user's email, and temporarily lock sign-in after repeated failed attempts. Users shall be able to reset their password through email. **Rationale:** Weak passwords and unlimited sign-in attempts let attackers reach users' private data, while legitimate users must still be able to regain access to their accounts. **Fit Criterion:** Passwords shorter than 8 characters or found on a common-password list shall be rejected. Sign-in to an account shall be locked for 15 minutes after 5 consecutive failed attempts. A user who enables multi-factor authentication shall be asked for an email code at each new sign-in, and a password reset email shall let a locked-out user sign in again. **Hazard Analysis:** SR1 (H1.1, H1.2)

ID	Requirement	Specification
SEC-ACC-05	Session End and Device Unlinking	**Description:** Flexin shall end a user's session after a period of inactivity or when the user signs out, and shall unlink the user's fitness devices on sign-out. **Rationale:** Sessions and device links left open on shared or lost devices let other people access an account or send their health data to it. **Fit Criterion:** A session unused for 30 days shall require the user to sign in again. After sign-out, the user's access and refresh tokens shall be rejected with HTTP 401, and no integration from that device shall be able to submit activity to the account. **Hazard Analysis:** SR2 (H1.2, H11.2)

15.2 Integrity Requirements

ID	Requirement	Specification
SEC-INT-01	Activity Validation	**Description:** The Spring backend shall validate activity metrics and detect duplicate records, overlapping timestamps, and implausible activity values before awarding virtual currency. **Rationale:** Validation prevents activity manipulation and unauthorized currency generation. **Fit Criterion:** Automated tests shall confirm that duplicate or flagged implausible activity records are marked unverified and receive zero virtual currency. Sustained cadence above 250 steps per minute shall be flagged for verification.

ID	Requirement	Specification
SEC-INT-02	Ledger Consistency	**Description:** All virtual currency operations, including rewards, trades, and dividends, shall use ACID-compliant MySQL transactions. **Rationale:** Transactional consistency prevents duplicate rewards, negative balances, and partial updates. **Fit Criterion:** Stress tests with 100 concurrent trade requests against one wallet shall verify that every operation commits consistently or rolls back completely, without partial state changes. **Hazard Analysis:** SR12 (H7.2, H9.2)
SEC-INT-03	Server-Side Valuation	**Description:** The Spring backend and custom market engine shall determine asset prices, portfolio valuations, and currency calculations. The mobile client shall not control authoritative values. **Rationale:** Server-side calculations prevent client manipulation of prices and balances. **Fit Criterion:** Trade requests attempting to specify unauthorized execution prices shall be rejected with HTTP 400.

ID	Requirement	Specification
SEC-INT-04	Data Backups	**Description:** Flexin shall keep regular backups of all user data, including accounts, activity history, balances, and portfolios. **Rationale:** Without backups, a database failure or bad write would leave users' balances, portfolios, and progress permanently lost. **Fit Criterion:** The production database shall be backed up at least once a day, and each backup shall be kept for at least 30 days. A restore from backup into a test environment shall be performed successfully before each release. **Hazard Analysis:** SR12 (H7.2, H9.2)

15.3 Privacy Requirements

ID	Requirement	Specification
SEC-PRV-01	Health Data Protection	**Description:** Sensitive personal information and fitness telemetry shall be protected through encryption in transit and at rest. **Rationale:** Encryption protects confidential user information against unauthorized disclosure. **Fit Criterion:** Client-server communications shall use TLS 1.3. Sensitive biometric data stored in MySQL shall be encrypted at rest using AES-256.

ID	Requirement	Specification
SEC-PRV-02	Private by Default	**Description:** All profile and health data shall be private by default. A field shall be shared with other users only after the user opts in, and backend access rules shall deny access to any data that has not been shared. Leaderboard participation shall be opt-in, and users may share activity streaks and simulated portfolio returns using a chosen alias. **Rationale:** Sensitive information such as weight, goals, and missed workouts must never be exposed to other users or unauthorized parties without consent. Users should control whether their activity and simulated financial performance are visible to others. **Fit Criterion:** All sharing settings, including leaderboard sharing, shall be disabled for new accounts. Enabling sharing shall require explicit consent and disclose which pseudonymous information and statistics will be shared. Automated access-rule tests run in CI shall confirm before each release that no user can read another user's unshared data. **Hazard Analysis:** SR3 (H2.1, H8.1, H9.1), SR9 (H4.1, H8.1)

ID	Requirement	Specification
SEC-PRV-03	User Data Deletion	**Description:** Users shall be able to request account deletion through the application settings, removing associated personal records. **Rationale:** Users should retain control over their personal information and account history. **Fit Criterion:** Personal health metrics, credentials, and fitness history shall be purged from the active database within 24 hours of a deletion request. Only appropriately anonymized transaction aggregates may be retained for auditing.

15.4 Audit Requirements

ID	Requirement	Specification
SEC-AUD-01	Economic Event Logging	**Description:** Flexin shall maintain an append-only audit ledger for currency rewards, simulated trades, dividends, and market price updates. **Rationale:** Auditable records support troubleshooting, transaction verification, and evaluation of the simulated economy. **Fit Criterion:** Database permissions shall prevent UPDATE and DELETE operations on audit records. Every recorded transaction shall reference a valid source activity or market event, where applicable.

ID	Requirement	Specification
SEC-AUD-02	Security Event Logging	**Description:** The backend shall log failed authentication, token validation failures, duplicate activity submissions, anti-exploit flags, and runtime exceptions. **Rationale:** Security logs help identify suspicious activity and diagnose integration or system failures. **Fit Criterion:** Structured logs shall capture severity, client IP address where available, authenticated user ID where applicable, and exception details for runtime errors.

15.5 Immunity Requirements

ID	Requirement	Specification
SEC-IMM-01	Injection Defense	**Description:** The backend and database layer shall be protected against SQL injection, cross-site scripting (XSS), and command injection. **Rationale:** Injection vulnerabilities could compromise database integrity and expose user information. **Fit Criterion:** Database access shall use parameterized queries or prepared statements. Static analysis and security reviews shall identify zero unresolved high-severity injection vulnerabilities before release.

ID	Requirement	Specification
SEC-IMM-02	Rate Limiting	**Description:** The Spring API shall limit request rates for authentication, trade submission, and activity synchronization endpoints. **Rationale:** Rate limiting reduces brute-force attempts, automated abuse, and service overload. **Fit Criterion:** A configured limit, initially 60 requests per minute per authenticated user or client IP, shall be enforced. Excess requests shall receive HTTP 429. Automated tests shall verify enforcement.
SEC-IMM-03	Client Environment Validation	**Description:** The Android client shall detect supported indicators of rooted or otherwise untrusted environments and restrict sensitive functionality when tampering is detected outside developer builds. **Rationale:** Compromised environments can increase the risk of credential exposure and activity manipulation. **Fit Criterion:** In supported production configurations, detected root or tampering indicators shall trigger restrictions on community leaderboard access and virtual currency minting. Developer builds shall remain available for authorized testing.

16 Cultural Requirements

This section outlines the cultural requirements for the software system. Cultural requirements refer to the expectations and norms of the target user base, which may vary based on geographic location, language, social norms, and other cultural factors. These requirements ensure that the software is culturally sensitive and appropriate for its intended audience.

16.1 Cultural Requirements

ID	Requirement	Specification
CUL-01	Supports Canadian Languages	**Description:** The system shall present all user-facing functionality with the option to choose between English and French. **Rationale:** The application will initially be deployed strictly within Canada, where English and French are the country's official languages. **Fit Criterion:** All user-facing text shall be stored separately from application logic and be replaceable through a localization framework.
CUL-02	Inclusive Financial Education	**Description:** The system shall provide financial education content that does not assume prior investing knowledge. **Rationale:** Users possess varying levels of financial literacy. The system seeks to educate users of all levels of financial literacy - including those who may have limited exposure to investing concepts. **Fit Criterion:** All investment-related terminology presented to users shall include a plain-language explanation accessible from within the application. All financial concepts shall be explained in simple, non-technical language to start, until the system deems the user has gained sufficient experience within the application. During user testing, a survey will be conducted to gather feedback on the clarity and understandibility of the financial education content - at least 80% of users shall express great satisfaction with little to no problems understanding financial concepts.

ID	Requirement	Specification
CUL-03	Inclusive Fitness Participation	**Description:** The system shall accommodate users with varying fitness levels by allowing personalized activity participation. **Rationale:** Users may differ significantly in age, physical ability, health conditions, and exercise experience. The application should promote participation regardless of fitness background. **Fit Criterion:** Users shall be able to participate in their preferred fitness activities while still gaining subsequent rewards.
CUL-04	Culturally Neutral Educational Content	**Description:** The system shall present fitness and financial education content using culturally neutral language and examples (i.e. sample companies in the virtual market). **Rationale:** The Canadian population consists of individuals from diverse cultural, religious, and socioeconomic backgrounds. Educational content should be inclusive and broadly applicable. **Fit Criterion:** Educational content shall not require familiarity with a specific cultural background, religion, ethnicity, or region to be understood.

17 Compliance Requirements

17.1 Legal Requirements

Insert your content here.

17.2 Standards Compliance Requirements

ID	Requirement	Specification
COMP-STD-01	Android Platform Compliance	**Description:** The system shall comply with the Google Play Developer Program Policies. **Rationale:** The application is intended to be deployed on the Google Play Store as to be accessible on Android devices and as a result must conform to platform requirements. **Fit Criterion:** The application shall successfully pass all mandatory checks required for distribution through the Google Play Store.
COMP-STD-02	Accessibility Compliance	**Description:** The system shall comply with WCAG 2.1 Level AA accessibility guidelines for user-facing functionality. **Rationale:** The system should offer a simple and delightful user experience that should be accessible to users with varying abilities (i.e. user's with colour-blindness, hearing impairments, etc.). **Fit Criterion:** All major user workflows shall satisfy WCAG 2.1 Level AA success criteria as evaluated using automated accessibility tools and manual testing.
COMP-STD-03	TypeScript Coding Standards Compliance	**Description:** The frontend codebase shall comply with the Google TypeScript Style Guide. **Rationale:** Consistent coding standards improve maintainability, readability, and long-term project quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.

ID	Requirement	Specification
COMP-STD-04	Java Coding Standards Compliance	**Description:** The backend codebase shall comply with the Oracle Java Code Conventions. **Rationale:** Standardized coding practices improve maintainability, code review effectiveness, and software quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.
COMP-STD-05	Database Compliance	**Description:** The system shall store data in compliance with the MySQL Reference Manual and the SQL Standard (ISO/IEC 9075). **Rationale:** The project architecture relies upon MySQL as the primary relational database system. **Fit Criterion:** All schema definitions and queries will be tested to execute successfully on the supported MySQL version without compatibility warnings or errors.
COMP-STD-06	Software Testing Standard Compliance	**Description:** The system shall undergo thorough and complete testing in accordance with the IEEE 29119 Software Testing Standard. **Rationale:** Consistent testing practices reduce defects and improve software reliability. The system will comply with this standard to provide a unified, process-driven framework for software testing. **Fit Criterion:** All code merged into the main branch shall pass automated test execution in the CI/CD pipeline based on tests that were developed in accordance with the IEEE 29119 Software Testing Standard.

ID	Requirement	Specification
COMP-STD-07	Information Security Compliance	**Description:** The system shall be designed in accordance with the security principles outlined in ISO/IEC 27001. **Rationale:** The application stores personal, financial-learning, and fitness-related information that must be protected against unauthorized access and disclosure. **Fit Criterion:** Security controls for authentication, access control, data storage, and data transmission shall align with ISO/IEC 27001 recommendations.

18 Open Issues

Insert your content here.

19 Off-the-Shelf Solutions

19.1 Ready-Made Products

The team looked at existing products that could meet the project's goals without building something new. Each covers part of the problem, but none connects fitness habits to learning how investments grow.

Stock market simulators (e.g. Investopedia Simulator): Free simulators give users virtual cash to trade real securities at market prices, and some let users compete in private groups. They teach investing well, but only to people who are already motivated to learn about it. The Problem Statement identifies young adults who are not yet interested as the group furthest behind, so a simulator alone does not reach them.

Step reward apps (e.g. Sweatcoin): These apps turn steps into a virtual currency that can be spent on offers and products. They reward activity, but the currency is spent rather than invested, so they teach nothing about compounding.

Move-to-earn crypto apps (e.g. STEPN): These reward exercise with cryptocurrency tokens that have real market value. They link exercise to money,

but expose users to real financial loss. STEPN's reward token lost most of its value within months in 2022. This approach conflicts with constraint SC5 (simulated market only) and is not suitable.

Fitness trackers (e.g. Strava, Garmin Connect, Health Connect): These record activity accurately but do not offer any financial component. Flexin will use them as data sources rather than replace them.

Since no ready-made product covers both halves of the problem, the team will build Flexin, reusing the components and ideas listed below.

19.2 Reusable Components

The following existing components are candidates for reuse. Each must have a license compatible with the project's proprietary license, and must be free to use within the project's budget.

Fitness data:

- `react-native-health-connect`, an open-source (MIT) React Native library for reading data from Android Health Connect.
- The Garmin Connect IQ SDK and Strava API for external devices and services.

Mobile client:

- React Navigation for moving between screens.
- An open-source React Native charting library for the portfolio graph and activity history. The specific library will be chosen during design.

Backend:

- Spring Boot to set up and run the API.
- Spring Security for authentication and access control, rather than writing account security from scratch.
- Spring Data JPA for reading and writing the MySQL database.

Market data: A source of historical or live asset prices for the simulated market. The team will choose it at the Market Decision milestone, and it must allow free use in an educational app.

Testing and tooling: JUnit, Detox, ESLint, and GitHub Actions, as listed in the Development Plan.

19.3 Products That Can Be Copied

The products below are not reused directly, but parts of their design can be adapted for Flexin. Only ideas will be copied, not code, artwork, or other protected material.

Investment simulators: The layout of portfolio, trade, and order history screens, which users of investing apps will already recognize. Short explanations of financial terms placed next to where the terms are used.

Step reward apps: Converting steps into currency at a fixed rate, with a daily limit on how much can be earned (see PR-SC1).

Strava: Leaderboards, clubs, and activity feeds that let friends follow each other's progress. These can be adapted for the community leaderboard and friend groups.

Habit and learning apps (e.g. Duolingo): Streaks, short daily goals, and reminders that build a daily habit. These must be adapted carefully so that missing a day is not punished (see PR-SC2).

20 New Problems

20.1 Effects on the Current Environment

Insert your content here.

20.2 Effects on the Installed Systems

Insert your content here.

20.3 Potential User Problems

Insert your content here.

20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Insert your content here.

20.5 Follow-Up Problems

Insert your content here.

21 Tasks

This section outlines the tasks that will be undertaken during the development of the solution. Each task is associated with a specific duration, start and finish date, and any dependencies on other tasks. The tasks are designed to ensure that the project progresses in a structured manner, with clear milestones and deliverables. This section also includes the phase-in plan identifying when requirements will be phased into the development process.

21.1 Project Planning

The team will follow an agile development process throughout the development of this solution, with a focus on iterative development and continuous feedback. The project will be divided into several sprints, each lasting two weeks. At the end of each sprint, the team will conduct a review to assess progress and make necessary adjustments. Each sprint will consist of planning, developing, testing, and deploying. This will allow the team to make iterative progress while still ensuring that previous iterations are effective and meet the requirements of the stakeholders. The detailed breakdown of weekly tasks and deliverables can be found in section 9 of the development plan and is the basis for the following project task breakdown:

Task	**Duration**	**Start**	**Finish**	**Dependency**
Requirements Gathering & Analysis	4 weeks	21-Sep-26	18-Oct-26	None
User Interviews & Validation	3 weeks	05-Oct-26	25-Oct-26	Requirements Gathering & Analysis
Fitness API Research & Integration Design	3 weeks	12-Oct-26	01-Nov-26	Requirements Gathering & Analysis
Virtual Currency System Design	2 weeks	26-Oct-26	08-Nov-26	User Interviews & Validation
Market Simulation Design	3 weeks	19-Oct-26	08-Nov-26	Requirements Gathering & Analysis

Task	Duration	Start	Finish	Dependency
UI/UX Design & Prototyping	4 weeks	26-Oct-26	22-Nov-26	User Interviews & Validation
Proof of Concept Development	5 weeks	02-Nov-26	06-Dec-26	Fitness API Design, Currency Design
Backend Architecture & Database Development	5 weeks	11-Jan-27	14-Feb-27	Design Document
Frontend Development	6 weeks	11-Jan-27	21-Feb-27	UI Design
Fitness Provider Integration	5 weeks	18-Jan-27	21-Feb-27	Backend Development
Portfolio Management Implementation	4 weeks	25-Jan-27	21-Feb-27	Backend Development
Gamification & Rewards Implementation	4 weeks	25-Jan-27	21-Feb-27	Currency Design
Rev 0 Demo Preparation	2 weeks	25-Jan-27	07-Feb-27	Core Features Complete
User Testing Round 1	2 weeks	01-Feb-27	14-Feb-27	Rev 0 Demo
Accessibility, Security, & Compliance Improvements	2 weeks	15-Feb-27	28-Feb-27	User Testing Round 1
Final User Testing	2 weeks	01-Mar-27	14-Mar-27	Accessibility, Security, & Compliance Improvements
V&V Execution	4 weeks	15-Feb-27	14-Mar-27	Most Development Tasks Complete
Final Bug Fixes & Stabilization	2 weeks	08-Mar-27	21-Mar-27	V&V Execution
Final Demo Preparation	1 week	15-Mar-27	21-Mar-27	Final Bug Fixes & Stabilization
EXPO Preparation	2 weeks	22-Mar-27	05-Apr-27	Final Demo Preparation

Table 28: Project Schedule and Task Breakdown

21.2 Planning of the Development Phases

Insert your content here.

22 Migration to the New Product

22.1 Requirements for Migration to the New Product

Insert your content here.

22.2 Data That Has to be Modified or Translated for the New System

Insert your content here.

23 Costs

Insert your content here.

24 User Documentation and Training

24.1 User Documentation Requirements

ID	Requirement	Specification
DOC-01	Financial Glossary and Tooltips	**Description:** Flexin shall provide contextual tooltips and a glossary defining financial concepts, including compound interest, inflation, ETFs, bond yields, and portfolio diversification. **Rationale:** Contextual explanations help novice users learn financial terminology while using the application. **Fit Criterion:** All financial instruments and portfolio metrics shall link to plain-language definitions. At least 90% of novice test participants shall locate and understand an ETF or compound return definition within 15 seconds.

ID	Requirement	Specification
DOC-02	Fitness Integration Guide	**Description:** Flexin shall provide an illustrated setup guide explaining how to authorize and configure supported fitness integrations and troubleshoot synchronization issues. **Rationale:** Clear instructions help users connect their fitness data sources successfully. **Fit Criterion:** Screenshot-assisted instructions shall cover all supported providers. At least 95% of test participants shall complete authorization using the guide alone.
DOC-03	Searchable Help Center	**Description:** Flexin shall provide an in-app searchable FAQ covering activity rewards, simulated market mechanics, virtual currency disclaimers, and health-data privacy. **Rationale:** Accessible documentation improves transparency, reduces confusion, and builds user trust. **Fit Criterion:** The help center shall operate without an external browser, support real-time keyword filtering, and return search results in under 200 milliseconds.
DOC-04	Digital Manual and Release Notes	**Description:** The team shall maintain a digital user manual and release notes in the project GitHub documentation repository. **Rationale:** Documentation enables supervisors, reviewers, and evaluators to verify functionality and assess project deliverables. **Fit Criterion:** A self-contained PDF and Markdown manual shall document architecture, navigation, data flows, and troubleshooting, with release notes for major project milestones.

Table 29: Flexin User Documentation Requirements

24.2 Training Requirements

Table 30: Flexin Training Requirements

ID	Requirement	Specification
TRN-01	Zero-Training Design	**Description:** Flexin shall follow a walk-up-and-use design requiring no mandatory formal training or external assistance. **Rationale:** Intuitive navigation reduces onboarding effort and encourages continued use. **Fit Criterion:** At least 90% of first-time test users shall register, connect a fitness data source, and view their dashboard within 3 minutes without assistance.
TRN-02	Interactive Onboarding	**Description:** After account creation, Flexin shall offer an optional, dismissible four-step walkthrough covering fitness goals, activity rewards, simulated investing, and the compounding chart. **Rationale:** Guided interaction introduces the application's core features through practical examples. **Fit Criterion:** The walkthrough shall take under 90 seconds, provide a visible "Skip Tour" option on every step, and remain accessible through settings.
TRN-03	Progressive Feature Discovery	**Description:** Flexin shall progressively introduce advanced features, including portfolio rebalancing, volatility filters, and leaderboard sharing, as users gain experience. **Rationale:** Gradual feature discovery reduces cognitive overload for novice users. **Fit Criterion:** Advanced trading indicators and portfolio analytics shall remain collapsed or secondary until the user completes an initial trade or logs at least three days of physical activity.

Continued on next page

Table 30: Flexin Training Requirements (Continued)

ID	Requirement	Specification
TRN-04	Demonstrator Quick-Start Kit	**Description:** The team shall provide a demonstrator guide and automated demo-data seeding script for project evaluations and presentations. **Rationale:** Preloaded data enables evaluators to verify application features without completing real-time workouts. **Fit Criterion:** A preset configuration or debug toggle shall load 30 days of simulated activity and portfolio history within 10 seconds, enabling demonstrations of trading and compounding charts.

25 Waiting Room

Insert your content here.

26 Ideas for Solution

Insert your content here.

27 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

27.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

27.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

27.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

28 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?